MILITARY SPECIFICATION

DRIFT METER, TYPE B-5

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers certain design requirements and all performance requirements for procurement of the type B-5 drift meter, which is used in the navigation of aircraft.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

2.1 The following publications, of the issue in effect on date of invitation for bids, shall form a part of this specification to the extent specified herein.

2.1.1 Specifications.-

Federal

QQ-A-354	Aluminum-alloy (AL-24) (aluminum-copper-magnesium (1.50%)-manganese); bars, rods, shapes, and wire.
QQ-A-591	Aluminum - Base - Alloy; Die Castings
QQ-P-416	Plating; Cadmium (Electrodeposited)

Military

MIL-D-5028	Drawings and Data Lists; Preparation of (For Engines, Accessories and Other Auxiliary Equipment)
MIL-E-5556	Enamel; Camouflage, Quick Drying
MIL-F-6775	Filter; Eyepiece
MIL-P-6064	Packaging of Lightweight Aircraft Accessories
MIL-P-6906	Plates, Information and Identification
MIL-P-6889	Primer; Zinc-Chromate, for Aircraft Use

Air Force-Navy Aeronautical

AN-QQ-A-696	Anodic-Films; Corrosion-Protective (for) Aluminum Alloys
AN-E-3	Enamel; Aircraft, Gloss
AN-S-126	Screw-Threads; Standard Aircraft

2.1.2 Other Publications.-

Air Force-Navy Aeronautical Bulletin

No. 143 Specifications and Standards; Use of

Bureau of Supplies and Accounts

Navy Shipment Marking Handbook

2.1.3 Drawings.-

Military Standards (Sheet)

MS28022 Drift Meter - Type B-5

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Air Force-Navy Aeronautical Design Standard Drawing

AND10398 Metals - Definition of Dissimilar

(Copies of this specification and copies of other publications referenced herein or required for Government procurement, and the Index of Military Aeronautical (AN or MIL) Standards, may be obtained upon application to the Commanding Officer, U. S. Naval Air Station, Johnsville, Pennsylvania.)

3. REQUIREMENTS

3.1 Materials.-

3.1.1 Metals.- Metals shall be of the corrosion-resisting type or suitably protected to resist corrosion in salt spray or atmospheric conditions to which the drift meter may be subjected when in storage or during normal service life.

3.1.1.1 Dissimilar Metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in Drawing AND10398.

3.1.2 Nonmagnetic Materials.- Nonmagnetic materials shall be used for all parts of the drift meter except where magnetic materials are essential.

3.1.3 Fungus-Proof Materials.- Materials which are not nutrients for fungi shall be used to the greatest extent practicable. Where materials which are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent, as approved by the Procuring Service.

3.1.4 Selection of Materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.1.4.1 AN or JAN Standard Parts.- AN or JAN Standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are replaceable by the AN or JAN Standard parts without alteration, and provided the corresponding AN or JAN part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding AN or JAN Standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.2 Design.- The outline dimensions of the drift meter shall conform to Standard MS28022. The drift meter shall be designed to include an optical system, a pantograph, a computer, channel rails, a gasket assembly and a mechanism frame and base.

3.3 Construction.- The drift meter shall be so constructed as to withstand the normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service without failure.

3.3.1 Optical System.- The optical system shall consist of an achromatic objective, field lens, reticle, and eye lens. The objective and field lens combined shall image a distant object, reversed and inverted, on the reticle. The field of view in the object space shall be not less than 35 degrees. The optical system shall be free from visible defects when looking through the drift meter, and shall be so adjusted that there shall be no parallax between the centers of the timing lines and the image of a distant object when viewed in the eyepiece.

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3.3.1.1 The focal lengths of the optics, dimensions of the reticle, and accuracy of mirror positioning shall be maintained within limits that permit ground speed measurement within an accuracy of ±2 percent.

3.3.1.2 Reticle.- A reticle, conforming to figure 1, that will rotate in unison with the drift index, shall be provided on the field lens in the periscope assembly of the drift meter.

3.3.1.3 Eye Lens.- The eye lens shall be so focused that a virtual image of a distant object shall be formed at a distance of between 1.1 and 2.6 meters (3.61 and 8.53 feet).

3.3.2 Pantograph.- The pantograph action shall be smooth and free. When the position of the pantograph is such that the links form a rectangle, the ball on the needle point shall lie within a 0.040-inch diameter circle, the center of which is the intersection of the line joining the fixed pivot to the center of the pencil holder and the center line of the pivots of the links carrying the needle.

3.3.2.1 Needle Point.- A needle, movable across the field of the eyepiece by means of a pencil through a pantograph linkage, shall be provided. The needle point shall be between 0.030 and 0.050 inch above the surface of the field lens carrying the reticle, when the pencil is centered on the recording plate and the pencil arm of the pantograph is centered in the guide slot.

3.3.2.1.1 When the center of the pencil holder is placed over the center of the grid plate, the ball on the needle point shall be approximately over the center line of the pivots of the links carrying the needle.

3.3.3 Measuring Mechanism.- The motion of the grid under the recording plate shall be smooth but sufficiently tight to prevent movement under vibration. For all settings of the drift index the drift lines of the grid on the reticle shall be parallel with those of the grid beneath the recording plate to within 15 minutes of arc.

3.3.3.1 Drift Scale.- The drift scale shall conform to figure 2.

3.3.3.2 Computer.- The computer shall conform to figure 3 and shall consist of a circular, rotatable, single-cycle logarithmatic scale concentric with a circular single-cycle logarithmatic fixed scale. The rotatable scale shall be graduated from 7 to 69, and the fixed scale shall be graduated from 50 to 490. A black reference line marked "MPH" and a red reference line marked "KNOTS" shall be located on the rotatable scale as shown on figure 3.

3.3.4 Channel Rails.- A channel-shaped rail shall be riveted on each side of the meter base plate for sliding the drift meter onto matching rails provided in the aircraft structure. Each rail shall be fitted with a tension spring and a detent to hold the drift meter in position.

3.3.5 Gasket Assembly.- The gasket assembly shall consist of a metal ring with two set screws on which a sponge rubber collar shall be mounted. The assembly shall seal the opening in the side of the aircraft through which the periscope tube projects.

3.3.6 Mechanism Frame and Base.- The mechanism frame and base shall be of cast aluminum conforming to Specification QQ-A-591, Class 2. The frame shall be provided with a hook for holding the pantograph pencil, and shall be hinged to the base.

3.4 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.5 Accessories.- Each drift meter shall be furnished with a filter adapter conforming to figure 4 and a type II, style I filter, conforming to Specification MIL-F-6775.

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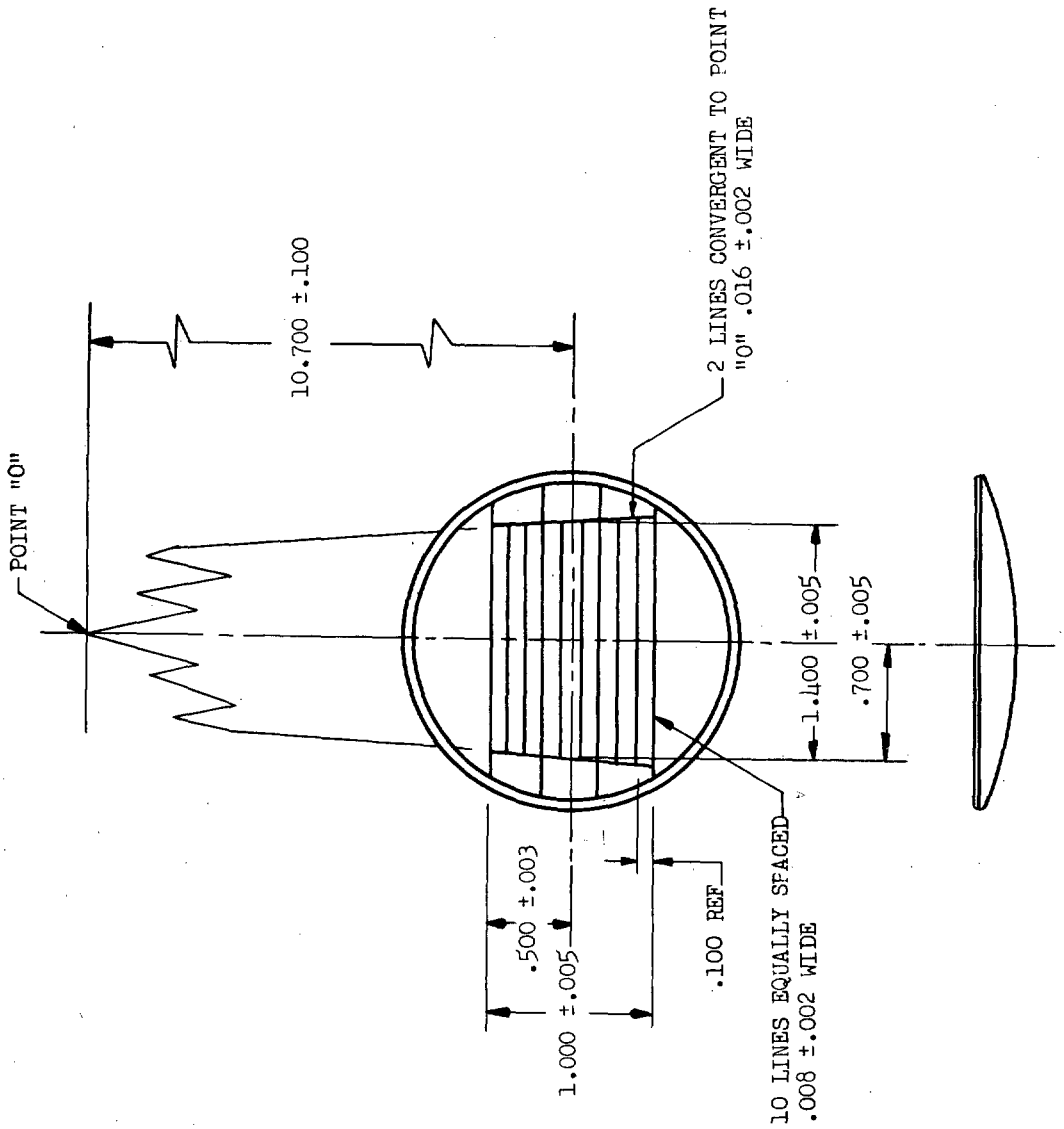
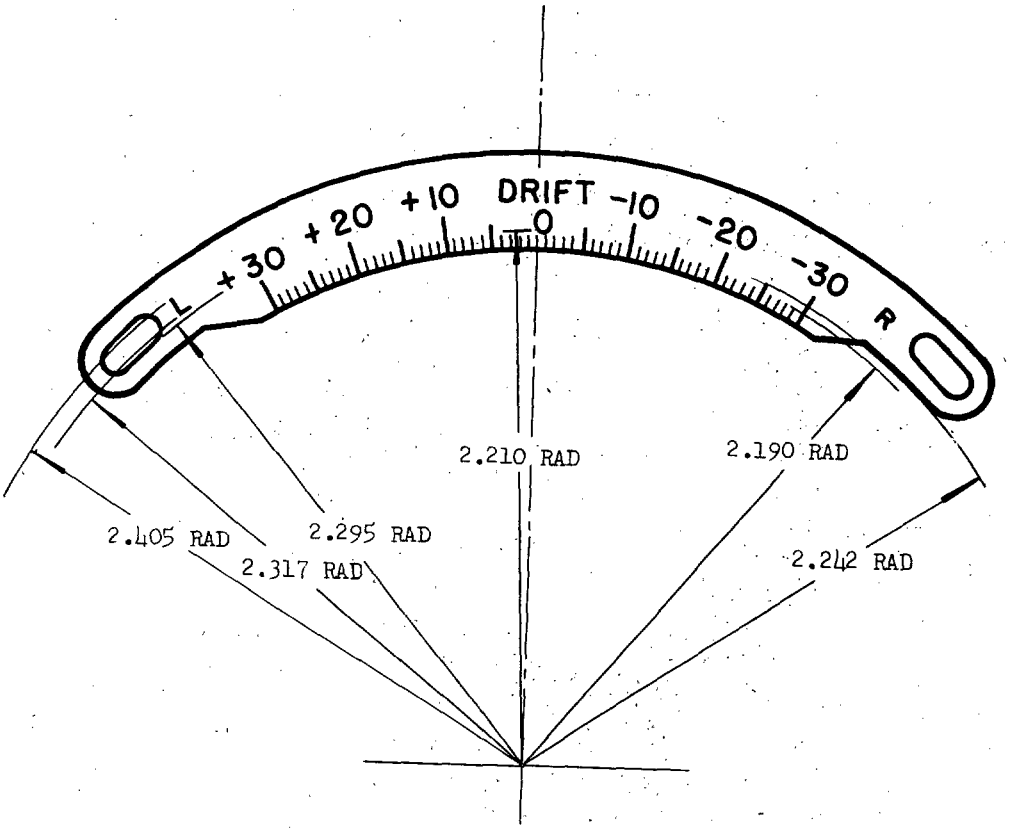


FIGURE 1. Reticle

DIMENSIONS IN INCHES.

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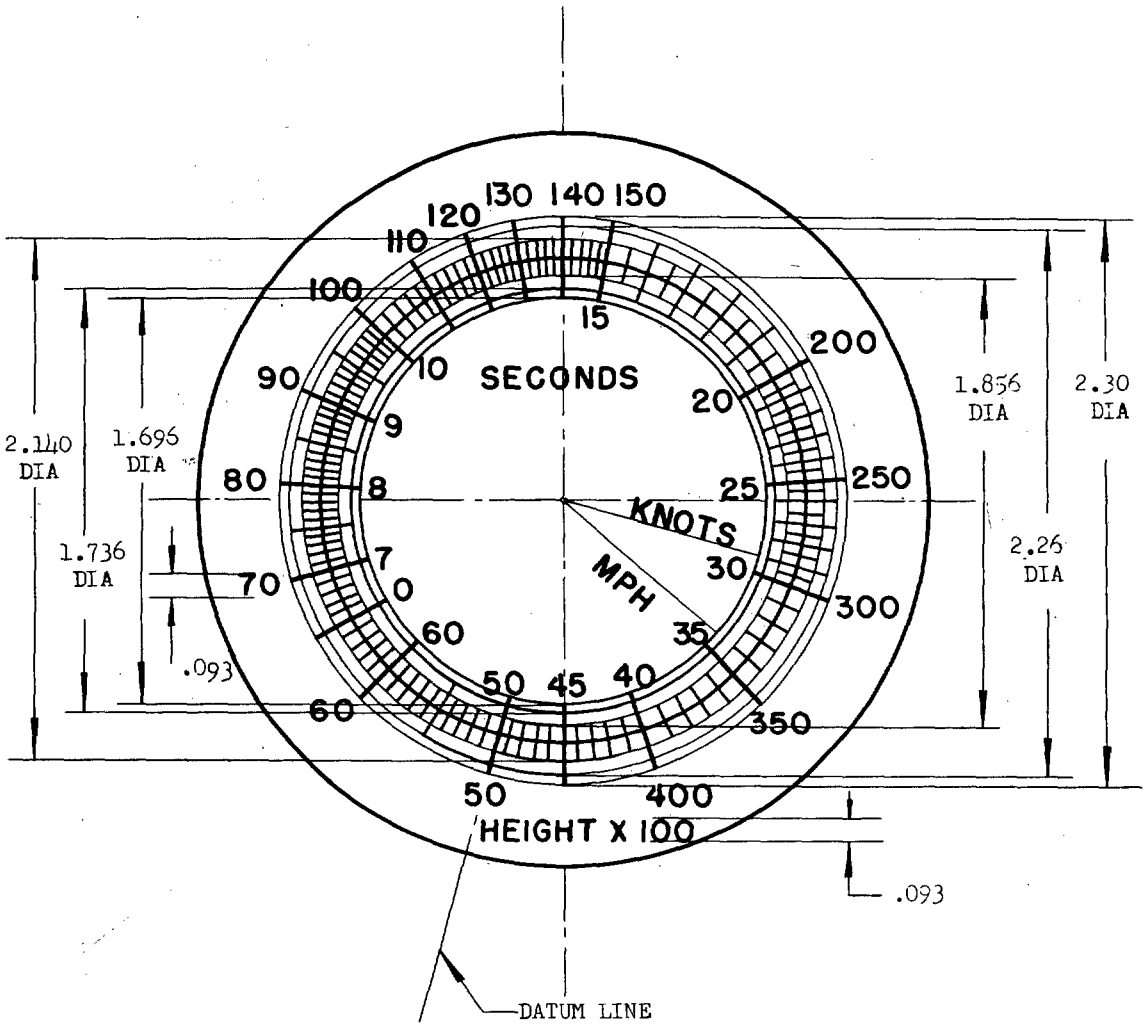


NOTE: FIGURES, LETTERS, LONG GRADUATIONS, + AND - SIGNS TO BE .015 WIDE.
SHORT GRADUATIONS TO BE .007 WIDE.

DIMENSION IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCE: DECIMALS ±.010.

FIGURE 2. Drift Scale

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NOTE: ALL LETTERS, FIGURES, AND LINES TO BE .010 WIDE.
DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCE: DECIMALS ±.010.

FIGURE 3. Computer

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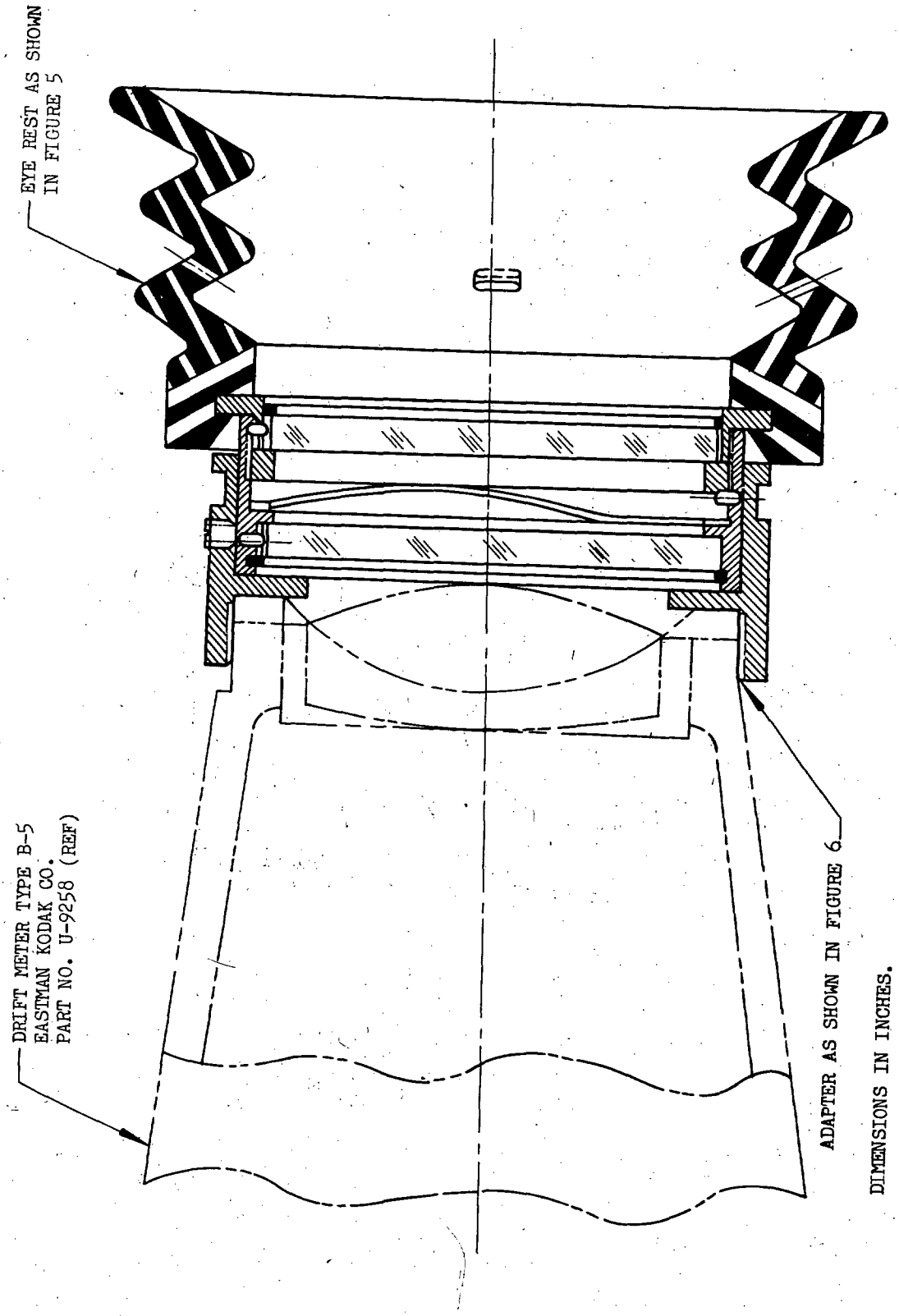


FIGURE 4. Adapter Assembly

Scale: Twice Size

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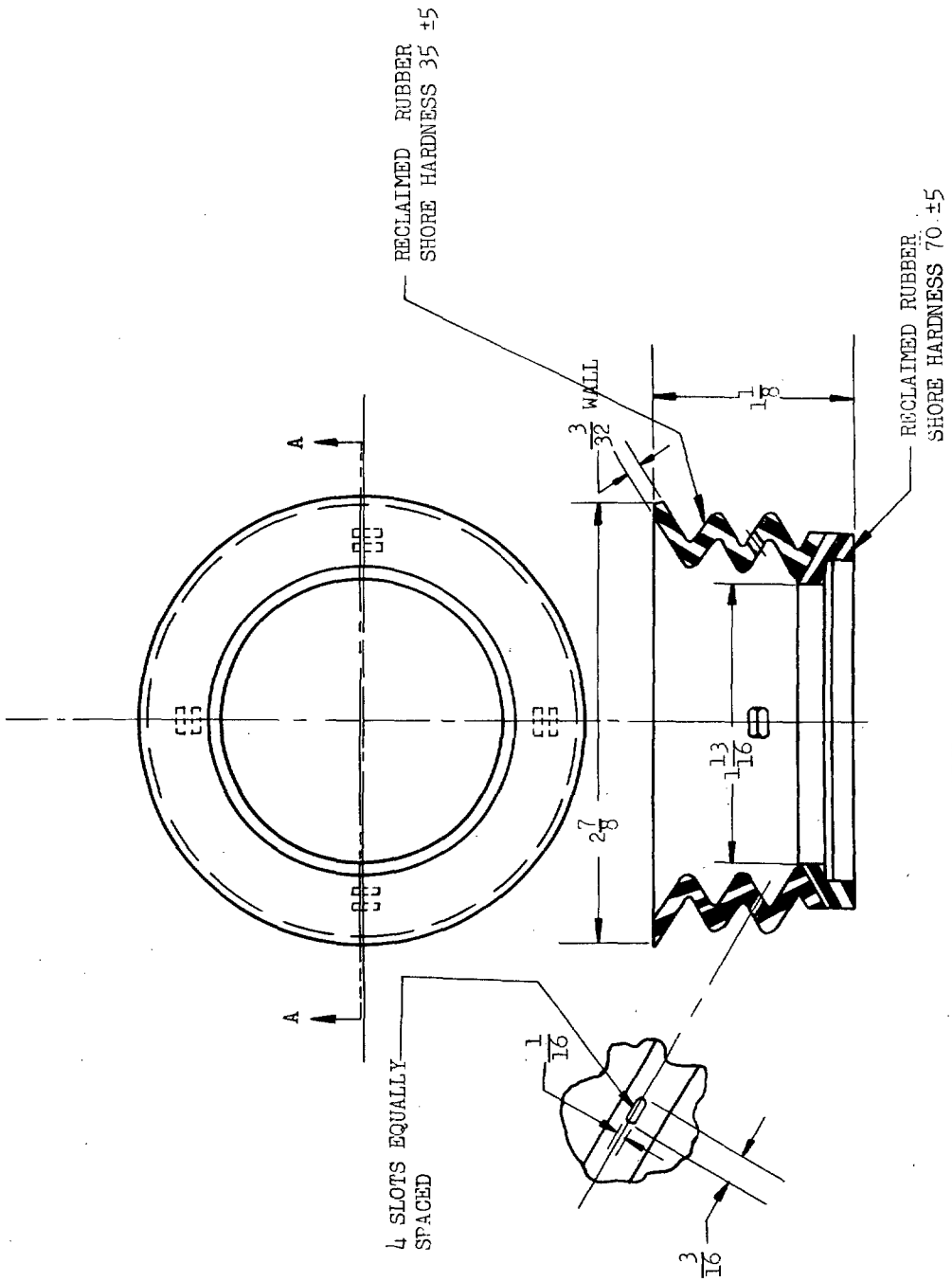
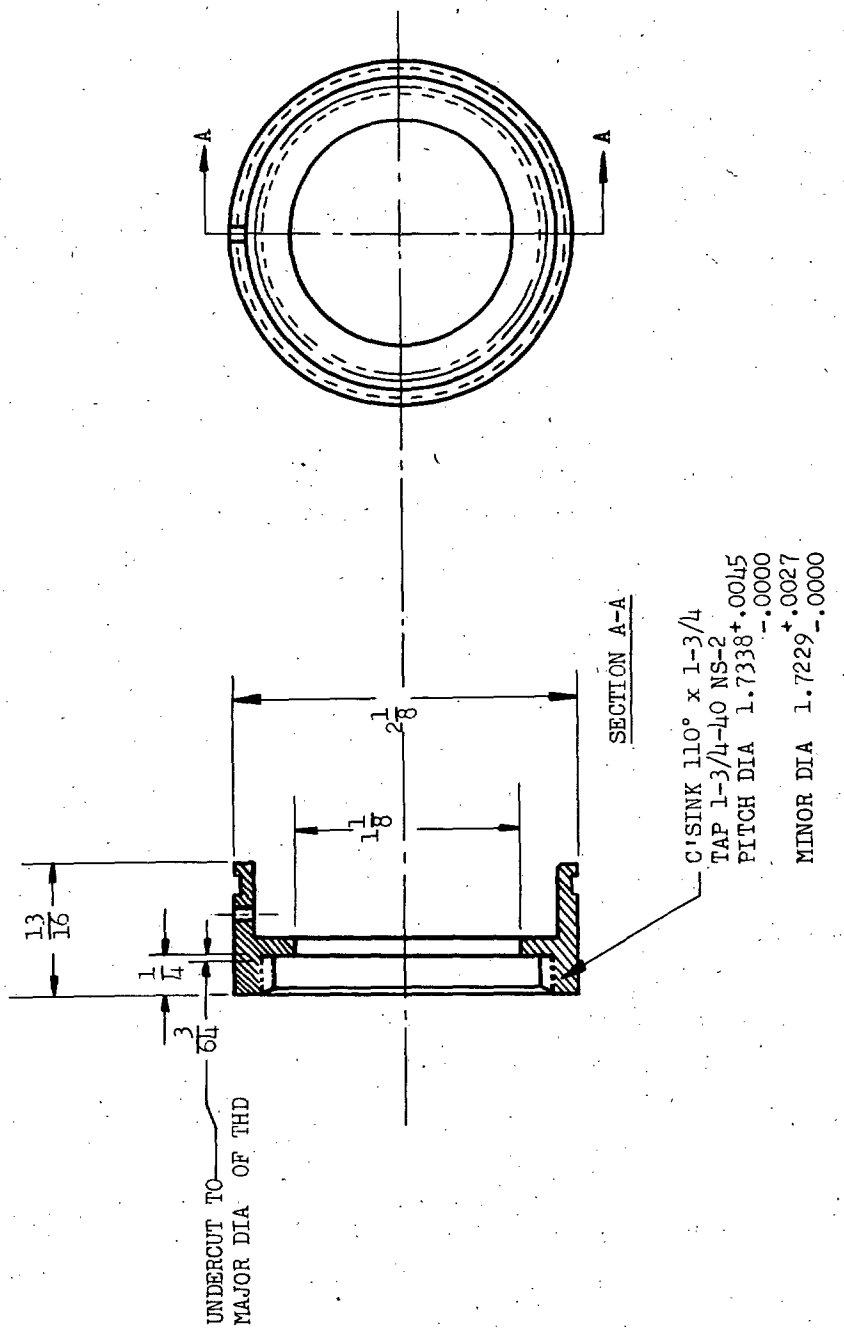


FIGURE 5. Eye Rest

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MATERIAL: ALUMINUM ALLOY, SPECIFICATION QQ-A-354.
FINISH: ANODIZE, SPECIFICATION AN-QQ-A-696; PRIMER AND ENAMEL - ONE THIN COAT OF ZINC-CHROMATE PRIMER CONFORMING TO SPECIFICATION MIL-P-6889, FOLLOWED BY TWO OR MORE COATS OF ENAMEL CONFORMING TO SPECIFICATIONS AN-E-3 OR MIL-E-5556. COLOR - BLACK

FIGURE 6. Adapter

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3.5.1 The manufacturer shall supply one spare recording plate with each drift meter.

3.6 Screw Threads.- Screw threads 0.060 inch or larger in diameter shall be in accordance with Specification AN-S-126.

3.7 Finish.- Protective coatings and finishes which will crack, chip, or scale during normal service life or due to extremes of atmospheric conditions shall not be used.

3.7.1 Aluminum-Alloy Parts.- Aluminum-alloy parts shall be covered with an anodic film conforming to Specification AN-QQ-A-696.

3.7.2 Steel Parts.- Steel parts shall be cadmium-plated in accordance with Specification QQ-P-416.

3.7.3 Optical System Housing.- The interior of the optical system housing shall be finished in dull black.

3.7.3.1 The external reflector shall be a second surfaced mirror adequately protected against corrosion and tarnish.

3.8 Performance.- The drift meter shall satisfy the performance requirements specified in Section 4 when subjected to the applicable tests.

3.9 Special Tools.- The design of the drift meter shall be such as to accommodate to the greatest possible extent disassembly, reassembly, and service maintenance by those tools and items of maintenance equipment which are normally available as commercial standards.

3.10 Markings.- All markings shall be durable to withstand usage encountered in service.

3.11 Identification of Products.-

3.11.1 Nameplate.- A nameplate conforming to Specification MIL-P-6906, legibly and permanently marked with the information listed below, shall be securely attached to the drift meter. The information marked in the spaces provided in the nameplate shall be in accordance with Specification AN-M-13.

DRIFT METER, TYPE B-5
MS Part No.
Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's Name or Trade-Mark
U S Property

3.12 Installation Instructions.- The contractor shall furnish with each instrument, one printed copy of instructions, with illustrations and diagrams, if necessary, covering the installation of the instrument. Prior to printing, two copies shall be furnished to the Procuring Service for approval. The instructions shall be printed on 8-1/2" by 11" or 17" by 11" paper.

3.12.1 Envelope.- An envelope furnished by the contractor containing installation instructions shall be packaged with each instrument and shall be marked with the following information:

"IMPORTANT
THIS ENVELOPE CONTAINS
INSTRUCTIONS"

3.13 Workmanship.- All details of workmanship shall be in accordance with high-grade aircraft manufacturing practice. Particular attention shall be given to neatness and thoroughness of assembling, and freedom of parts from burrs and sharp edges.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Classification of Tests.- The inspection and testing of drift meters shall be classified as follows:

(a) Preproduction tests: Preproduction tests are those tests accomplished on samples which are representative of the production of the item after award of contract, to determine that the production meets the requirements of this specification.

(b) Inspection tests: Inspection tests are those tests accomplished on drift meters manufactured and submitted for acceptance under contract.

4.2 Test Conditions.-

4.2.1 Standard Atmospheric Conditions.- Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of approximately 29.92 inches of mercury and a temperature of approximately 25°C (77°F).

4.3 Preproduction Tests.-

4.3.1 Sampling Instructions.- The Preproduction test samples shall consist of two complete drift meters of each manufacturer's part number. The preproduction samples shall be submitted for test and evaluation to the agency designated in the Letter of Authorization from the Procuring Service. (See paragraph 6.3.1.)

4.3.2 Tests.- The preproduction sample tests shall consist of all the tests specified under Inspection tests.

4.4 Inspection Tests.- The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the Procuring Service. The contractor shall furnish test reports, in duplicate, showing quantitative results for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.4.1 Individual Tests.- Each drift meter submitted for acceptance under the contract shall be subjected to the following tests:

4.4.1.1 Examination of Product.- Each drift meter shall be examined externally to determine conformance with this specification with respect to selection of materials, workmanship, and all other requirements not covered by tests. All moving parts shall be checked to determine that they operate freely and accurately without excessive backlash.

4.4.1.2 Parallax.- The optical system shall be checked by viewing a distant object through the eyepiece. There shall be no parallax between the center of either timing line and the image of the distant object.

4.4.1.3 Clearance.- The clearance between the needle point and the field lens carrying the reticle shall be between 0.030" and 0.050" when the pencil is centered on the recording plate and the pencil arm of the pantograph is centered in the guide slot.

4.4.1.4 Pantograph Accuracy.- One of the inner lines of the parallel grid of the reticle shall be followed by the needle point. The line traced on the recording plate shall be parallel with the lines on the grid beneath, within 30 minutes of arc.

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4.4.1.5 Computer Accuracy.-

4.4.1.5.1 The line marked "10" on the rotatable scale shall be set opposite the line marked "100" on the fixed scale. All the lines on the rotatable scale shall coincide with those of the fixed scale to within 0.010 inch.

4.4.1.5.2 The line marked "10" on the rotatable scale shall be set opposite the line marked "200" on the fixed scale The line marked "30" on the rotatable scale shall coincide with the line marked "60" on the fixed scale to within 0.010 inch.

4.4.2 Sampling Tests.- The Sampling tests shall consist of the following tests conducted on one drift meter from every 100 drift meters, or less, submitted for acceptance under contract. These tests shall be in addition to the Individual tests specified herein.

4.4.2.1 Low Temperature Operation.- The drift meter shall be subjected to a temperature of -55° ±2°C (-67° ±3.6°F) for a period of 4 hours. After this period and with the drift meter maintained at this temperature, the meter shall be subjected to and shall meet the requirements specified for the Individual tests.

4.4.2.2 High Temperature Operation.- The procedure outlined for the "Low Temperature Operation" test shall be repeated, except the ambient temperature shall be +70° ±2°C (+158° ±3.6°F).

4.4.3 Rejection and Retests.- Any drift meter which has been rejected may be reworked or replaced to correct the defects, and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Drift meters rejected after retest shall not be resubmitted without specific approval of the Procuring Service.

5. PREPARATION FOR DELIVERY

5.1 Application.- The packaging, packing, and marking requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Preservation and Packaging.- The preservation and packaging shall be in accordance with Specification MIL-P-6064, Group VIII.

5.3 Packing.- Packing shall be in accordance with Specification MIL-P-6064.

5.4 Marking and Labeling.- The marking shall be in accordance with the requirements of Specification MIL-P-6064 and as specified herein.

5.4.1 Packages.- Each package shall be durably and legibly marked with the following information in such a manner that the markings will not become damaged when the packages are opened:

DRIFT METER, TYPE B-5
MS Part No.
Stock No.
Quantity
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's Name or Trade-Mark
U S Property

"IF IN STORAGE AFTER (Date)*, THE DRIFT METER SHALL BE TESTED AND INSPECTED BEFORE ISSUE."

*NOTE: This date shall be specified by the Procuring Service and shall be inserted by the Inspector at the time of acceptance.

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5.4.1.1 Acceptance Stamp.- Each package shall be marked with the Inspector's acceptance stamp, imprinted partly on the sealing strip and partly on the carton.

5.4.1.2 The package, in addition to other markings specified herein or by the Procuring Service shall be labeled or have the following imprinted on the exterior:

"This package has been prepared for overseas shipment and/or long time storage and shall not be opened until ready for use or for inspection. If opened, reprocessing as specified in Specification MIL-P-6064, Group VIII is necessary before being replaced in storage or being shipped."

5.4.2 Shipping Containers.- Shipping containers shall be marked as specified under paragraph 5.4.1, marked to indicate the presence and/or method of preservation and packaging, method of exterior packing, i.e., domestic or overseas. In addition, each shipping container shall be marked in accordance with the requirements of the Navy Shipment Marking Handbook.

5.4.3 Precautionary Markings.- The following marking shall appear on two opposite sides of each package and shipping container:

"FRAGILE
DELICATE INSTRUMENT, HANDLE WITH CARE"

6. NOTES

6.1 Intended Use.- The drift meters covered by this specification are intended for use by the navigator on aircraft for measuring the angle between the heading of the aircraft and its actual path over the ground, and to determine the ground speed of the aircraft in knots or miles per hour when the height of the aircraft is known.

6.2 Drawing Requirements.- The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of Specification MIL-D-5028.

6.3 Ordering Data.- Requisitions, contracts, and orders should specify the MS part number of the drift meter desired, the quantity, and whether overseas packing is to be furnished.

6.3.1 Provisions for Preproduction Tests.- The acceptance of the remaining drift meters on a contract or purchase order shall be dependent upon approval of the test results on the Preproduction samples required by paragraph 4.3.1 of this specification. The attention of manufacturers is called to this provision, and they are urged to submit the Preproduction samples at the earliest practicable date after the date of contract to the Navy. Information as to the marking and forwarding of samples should be addressed to the Bureau of Aeronautics, Navy Department, Washington 25, D. C.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Navy - Bureau of Aeronautics

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